

Immunophenotyping - Fig5

Léa ROGUE

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Load libraries

```
library(ComplexHeatmap)
library(dplyr)
library(tidyr)
library(ggplot2)
library(colorRamp2)
library(plotly)
library(ggpubr)
library(WGCNA)
```

```
# Function to calculate Z scores
calculate_z_scores <- function(df_input, col) {
  # Exclude col PROBEID SYMBOL and CTA
  data_values <- df_input[, -c(col)]

  # Calculate Z-scores
  z_scores_row <- t(scale(t(data_values)))

  # Add columns
  df_z_scores <- cbind(df_input[, c(col)], z_scores_row)

  # Return df
  return(df_z_scores)
}
```

Load and prepare data

```
# Read immunogenicity table
df_imgg <- read.table("../data/peptide_table_imgg_pred.tsv",
  sep = "\t", header = T)
df_imgg <- df_imgg[df_imgg$allele %in% "HLA-A*02:01", ]

# Read CTA peptides
df_cta_pep <- read.table("../results/peptides_CTA_uniques_jj_ch_iedb.tsv",
  sep = "\t", header = T)
colnames(df_cta_pep) <- c("Gene", "peptide")
df_imgg_genes <- merge(df_imgg, df_cta_pep, by = "peptide", all.x = T)

# Prepare scores df
df_gene_score <- df_imgg_genes %>%
  group_by(Gene) %>%
  summarise(mean_score = mean(immunogenicity.score, na.rm = TRUE),
    sum_score = sum(immunogenicity.score, na.rm = TRUE),
    n_peptides = n()) %>%
  arrange(desc(sum_score))

# Read CTAs
l_cta <- read.table("../data/CTA_list_clean.txt", header = F)$V1
```

```

# Matrix with intensities and information on immune
# signature and CTA
df_CTA_immune_whole_clean_avg_102 <- read.table("../results/whole_gene_int_CTA_sign_imm_clean.tsv",
  sep = "\t", header = TRUE, check.names = FALSE, row.names = 1)

# Metadata
df_metadata <- read.table("../results/metadata.tsv", sep = "\t",
  header = TRUE, check.names = FALSE, dec = ",")

# Conventional CHS metadata
df_metadata_conv <- df_metadata[df_metadata$Histology != "N/A" &
  df_metadata$Histology != "benign" & df_metadata$Histology !=
  "dedifferentiated", ]
df_metadata_surv_conv <- df_metadata_conv[, c("Patient", "OS.delay",
  "OS.event")]
df_metadata_surv_conv <- na.omit(df_metadata_surv_conv)

# Matrix for conventional patients with survival metadata
df_CTA_immune_whole_clean_avg_63 <- df_CTA_immune_whole_clean_avg_102[,
  c("Signature", "CTA", colnames(df_CTA_immune_whole_clean_avg_102)[colnames(df_CTA_immune_whole_clean_avg_102) != "Patient"])
  df_metadata_surv_conv$Patient]]
df_z_scores_63 <- calculate_z_scores(df_CTA_immune_whole_clean_avg_63,
  c(1, 2))

# Df for immune signatures
df_complete <- read.table("../results/matrix_complete_intensities.tsv",
  sep = "\t", header = TRUE, check.names = FALSE)
df_complete_63 <- df_complete[, c("Signature", "CTA", colnames(df_complete)[colnames(df_complete) != "Patient"])
  df_metadata_surv_conv$Patient]]

# Average the expression between same immune cells types
# Take rows with immune cells signature from normalized
# data
df_avg_immune_sign_63 <- df_complete_63 %>%
  filter(Signature != "NA")

# Group by signature and calculate mean of expression
# values
df_avg_immune_sign_final_63 <- as.data.frame(df_avg_immune_sign_63 %>%
  dplyr::select(-c(CTA)) %>%
  group_by(Signature) %>%
  summarise(across(where(is.numeric), \(x) mean(x,
    na.rm = TRUE))))
rownames(df_avg_immune_sign_final_63) <- df_avg_immune_sign_final_63$Signature
df_avg_immune_sign_final_63 <- df_avg_immune_sign_final_63[,
  -1]

# Compute z scores
df_imm_z_scores_63 <- t(scale(t(df_avg_immune_sign_final_63)))

# read scatter data
df_scatter_hr <- read.table("../results/table_scatter_63_complete.tsv",
  sep = "\t", header = T)

```

I. Correlation heatmap

```
data <- subset(df_z_scores_63, df_z_scores_63$CTA == "CTA")
data <- data[, -c(1, 2)]

# Pearson correlation and p-value compute
cor_matrix <- cor(t(df_imm_z_scores_63), t(data), method = "pearson")
cor_pval <- corPvalueStudent(cor_matrix, 63)

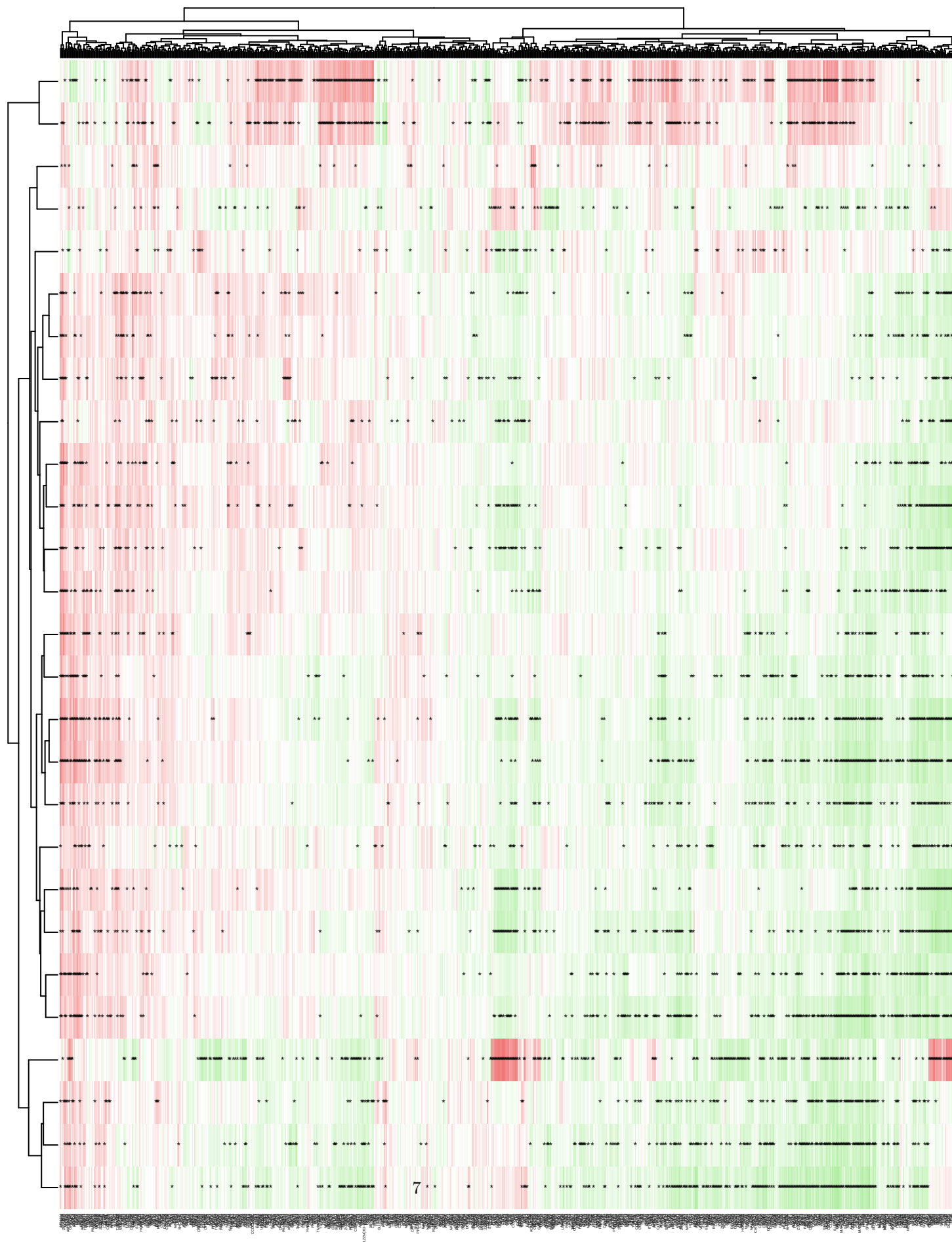
# Significant p-val
pval_signif <- cor_pval < 0.05

# Heatmap
colors <- colorRampPalette(c("#75E05A", "white", "#EA4343"))(100)
heatmap_all_cta_corr_63 <- Heatmap(as.matrix(cor_matrix), cluster_rows = TRUE,
  cluster_columns = TRUE, cluster_column_slices = TRUE, clustering_distance_columns = "euclidean",
  clustering_method_columns = "complete", show_column_dend = TRUE,
  column_names_gp = gpar(fontsize = 2), row_names_gp = gpar(fontsize = 5),
  col = colorRamp2(seq(-1, 1, length.out = 100), colors), heatmap_legend_param = list(title = "Pearson"),
  cell_fun = function(j, i, x, y, width, height, fill) {
    if (pval_signif[i, j]) {
      grid.text("*", x, y, gp = gpar(fontsize = 5, col = "black"))
    }
  })

## Warning: You defined `cell_fun` for a heatmap with more than 100 rows or
## columns, which might be very slow to draw. Consider to use the
## vectorized version `layer_fun`.

# pdf('../results/correlation_matrix_all_CTA_63_imgg.pdf',
# height = 6, width = 40)
heatmap_all_cta_corr_63 <- draw(heatmap_all_cta_corr_63)

# dev.off()
```



```

# Store clusters HOT and COLD
cta_clust <- column_order(heatmap_all_cta_corr_63)

# Create table with cta
df_immp <- data.frame(col_clusters = c(rep("Hot tumor CTAs",
length(cta_clust[1:402])), rep("Cold tumor CTAs", length(cta_clust[403:841]))),
  Gene = c(rownames(data)[cta_clust[1:402]], rownames(data)[cta_clust[403:841]]))

# Save write.table(df_immp, file =
# '../results/clusters_cta_pearson_63.tsv', sep = '\t',
# quote = FALSE, row.names = FALSE)

```


II. Scatter plot

```
# Merge immunophenotype and immunogenicity
df_imgg_imgp <- merge(df_gene_score, df_imgp, by = "Gene", all = T)
df_imgg_imgp <- df_imgg_imgp %>%
  filter(!is.na(col_clusters))

# Prepare data for scatter
df_scatter_h_c <- merge(df_scatter_hr, df_imgg_imgp, by.x = "SYMBOL",
  by.y = "Gene", all.x = T)
df_scatter_h_c$HR <- ifelse(is.na(df_scatter_h_c$HR), 0.1, df_scatter_h_c$HR)

# Plot
p <- ggplot(df_scatter_h_c, aes(x = Mean_expression_tissues,
  y = Intensity, color = Cluster, size = HR, shape = col_clusters)) +
  geom_point(alpha = 0.8) + geom_hline(yintercept = median(df_scatter_h_c$Intensity),
  linetype = "dashed", color = "black") + annotate("text",
  x = max(df_scatter_h_c$Mean_expression_tissues), y = median(df_scatter_h_c$Intensity) +
    0.3, label = "Median\nexpression", size = 2.5) + geom_vline(xintercept = quantile(df_scatter_h_c$Mean_expression_tissues,
  0.75), linetype = "dashed", color = "black") + annotate("text",
  x = quantile(df_scatter_h_c$Mean_expression_tissues, 0.75) +
    3.5, y = max(df_scatter_h_c$Intensity) + 0.2, label = "75th percentile",
  size = 2.5) + scale_size_continuous(range = c(1, 4), breaks = c(min(df_scatter_hr[!is.na(df_scatter_hr$HR)], 1, max(df_scatter_hr$HR, na.rm = TRUE)), labels = ~round(.x,
  2)) + scale_shape_manual(values = c(`Cold tumor CTAs` = 15,
  `Hot tumor CTAs` = 16)) + scale_color_manual(values = c(A = "#3498DB",
  B = "#2ECC71", C = "#E74C3C"), na.value = "#b0b0b0") + theme_minimal() +
  scale_x_log10() + labs(x = "CTAs expression in normal tissues (Log10 TPM mean)",
  y = "CTAs expression in CHS (mean intensities)", shape = "",
  size = "Sized according to HR\n(only for colored points)",
  color = "CTAs cluster")
p

# ggsave('../results/scatter_plot.svg', plot = p, width =
# 20, height = 15, units = 'cm')
```

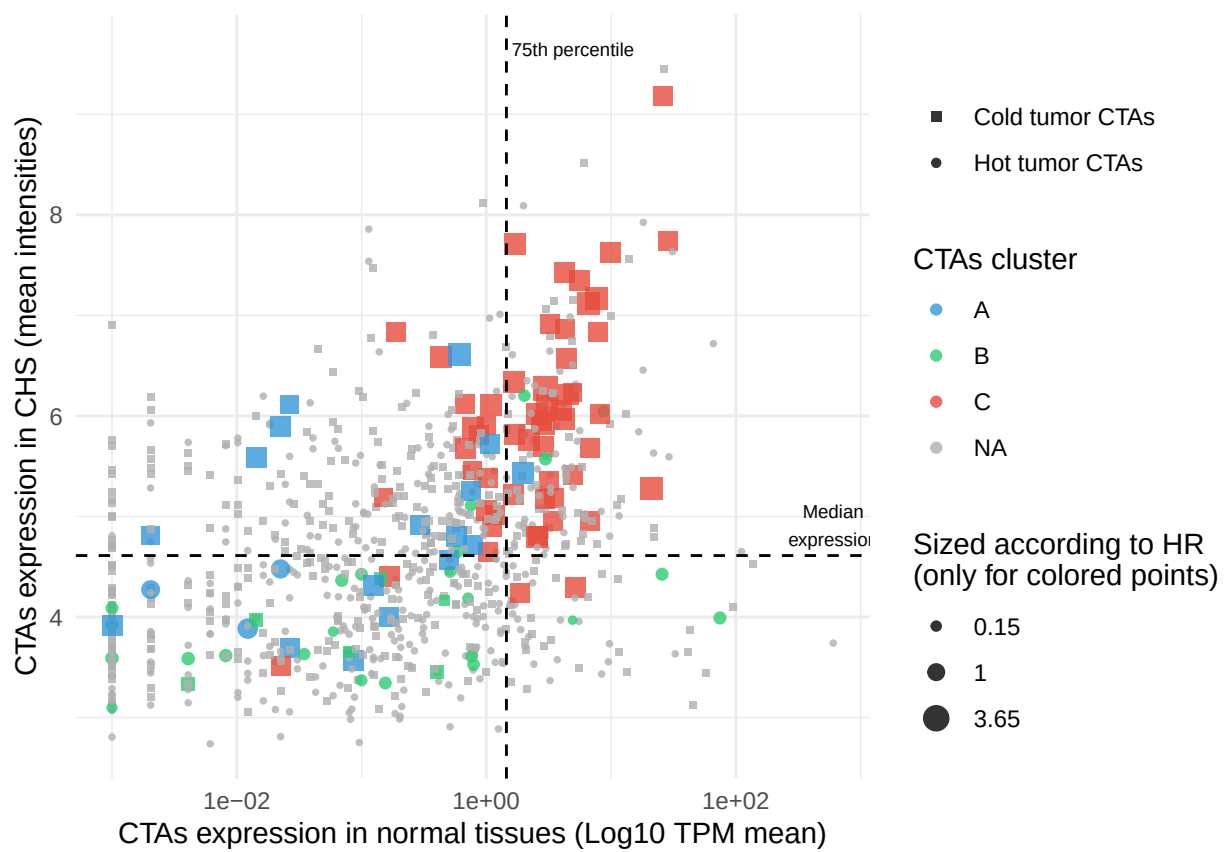


Figure 2: Scatter plot with immunophenotype data

III. Bar plot of CTAs number per cluster with stats

```
# Count
df_clean <- df_scatter_h_c %>%
  filter(!is.na(Cluster))
df_count <- df_clean %>%
  count(Cluster, col_clusters)

# Create table
tab <- df_clean %>%
  count(Cluster, col_clusters) %>%
  pivot_wider(names_from = col_clusters, values_from = n, values_fill = 0)

# Fisher pairwise test
pairwise_test <- function(c1, c2, data) {
  sub <- data %>%
    filter(Cluster %in% c(c1, c2))
  mat <- as.matrix(sub[, c("Hot tumor CTAs", "Cold tumor CTAs")])
  rownames(mat) <- sub$Cluster
  fisher.test(mat)$p.value
}

pairs <- combn(unique(tab$Cluster), 2, simplify = FALSE)

results <- data.frame(group1 = sapply(pairs, function(x) x[1]),
  group2 = sapply(pairs, function(x) x[2]), p = sapply(pairs,
    function(x) pairwise_test(x[1], x[2], tab)))

# correction multiple tests
results$p_adj <- p.adjust(results$p, method = "BH")

# Add stars
results$p_adj.signif <- cut(results$p_adj, breaks = c(-Inf, 0.001,
  0.01, 0.05, Inf), labels = c("***", "**", "*", "ns"))

# Annotation
totals <- df_count %>%
  group_by(Cluster) %>%
  summarise(total = sum(n))

ymax <- max(totals$total)

results$y.position <- c(ymax * 1.05, ymax * 1.15, ymax * 1.25)

# Barplot
p <- ggplot(df_count, aes(x = Cluster, y = n, fill = col_clusters)) +
  geom_bar(stat = "identity", position = "stack", width = 0.4) +
  scale_fill_manual(values = c(`Hot tumor CTAs` = "#9e9e9eff",
    `Cold tumor CTAs` = "#625f6aff")) + labs(x = "Cluster",
    y = "CTAs number", fill = "") + theme_classic()

# Add stars on the plot
```

```

plot_final <- p + stat_pvalue_manual(results, label = "p.adj.signif",
  xmin = "group1", xmax = "group2", y.position = "y.position",
  tip.length = 0.01, size = 6) + annotate("text", x = Inf,
  y = Inf, label = "* p < 0.05\n** p < 0.01\n*** p < 0.001",
  hjust = 1.1, vjust = 1.1, size = 4)
plot_final

```

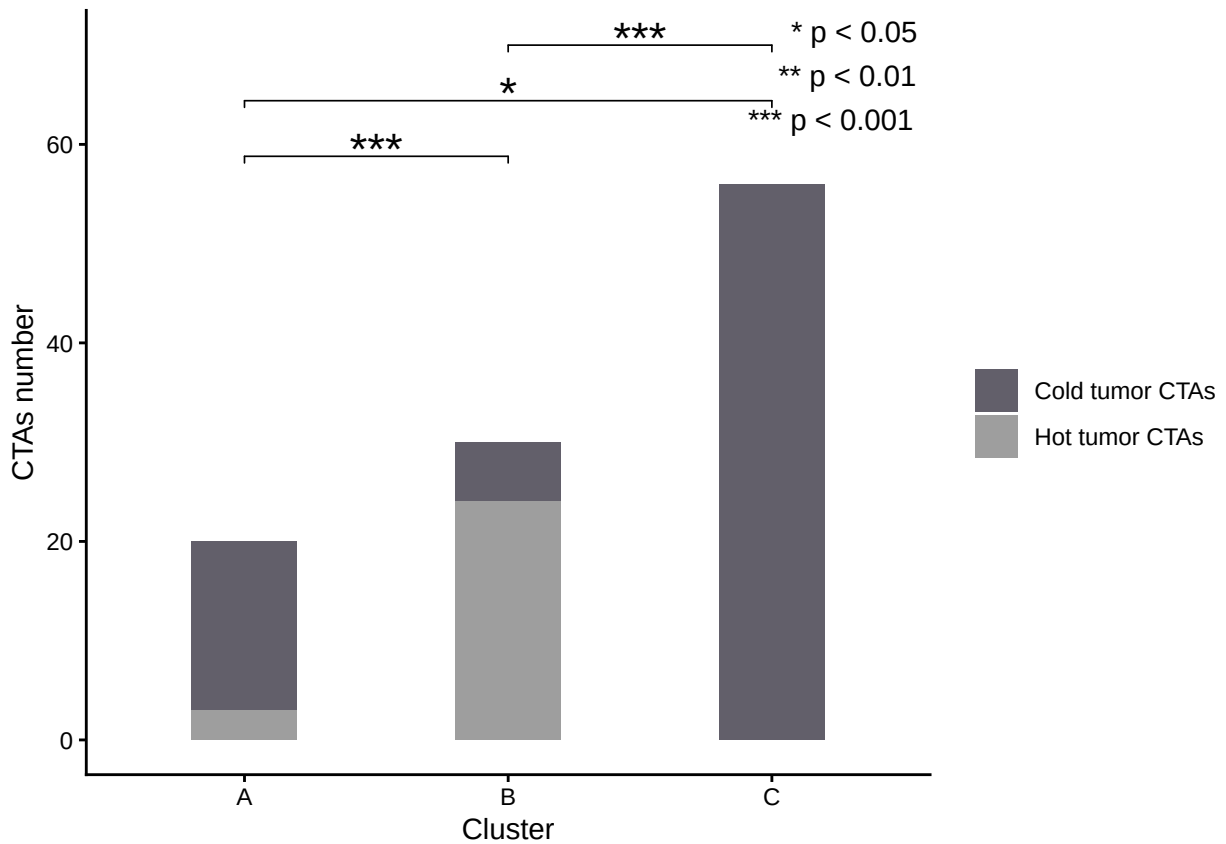


Figure 3: Bar plot of CTAs number per cluster with stats

```

# ggsave('../results/barplot.svg', plot = plot_final, width
# = 10, height = 10, units = 'cm')

```